

# Jamie Seoh

[j2seoh@uwaterloo.ca](mailto:j2seoh@uwaterloo.ca) | [linkedin.com/in/jamie-seoh](https://linkedin.com/in/jamie-seoh) | [github.com/dwseoh](https://github.com/dwseoh) | [dwseoh.com/](https://dwseoh.com/)

## EDUCATION

### University of Waterloo

*Bachelor of Software Engineering*

Sep 2025 – Present

3.95/4.00 (91.2%) GPA

- Top 7.6% in Euclid Math Contest, 2 x Hackathon Winner, Software Engineering Entrance Scholarship

## EXPERIENCE

### Cerebras Systems

*AI Software Engineer Intern* | C++, Python, Assembly, Bash, GTest, Pytest

May 2026 – Aug 2026

Toronto, Ontario

- Engineered an end-to-end **hardware fault-injection framework** that simulates crashes, stalls, and slowdowns across the inference pipeline, helping engineers reproduce and diagnose difficult-to-isolate customer failures.
- Reduced false-positive on-call alerts by **84%** by designing a two-tier hardware-fault detection system.
- Built a **KV-cache breakpoint mechanism** that pauses faulted workloads and captures diagnostics for analysis.
- Sped up developer productivity with a static linter for Cerebras assembly language and an interactive dashboard.

### SolveXchange

*Front-End Developer* | HTML, CSS, JavaScript, Flask, SQLAlchemy

Sep 2023 – Jan 2024

Online

- Shipped responsive frontend interfaces serving **100+ active users** on a nonprofit entrepreneurship platform.
- Implemented **JWT**-based authentication and REST APIs with **Flask** for user and forum workflows.

## PROJECTS

### Self-Playing MIDI Keyboard | C++, Arduino, Teensy 4.0, Hall-effect Sensors, Solenoids, 3D Printing

- Designed a 25-key MIDI keyboard with 3D-printed keys, velocity sensing, and USB-MIDI/synthesizer modes.
- Engineered a **Hall-effect sensing system** using multiplexed analog inputs, reducing pin usage from 25 to 6.
- Developed Teensy firmware for sensor processing, MIDI handling, velocity detection, and real-time key control.
- Built a **self-playing system** using solenoids, MOSFET switching, and PWM for independent key actuation.

### Quota — DeltaHacks12 First Place | TypeScript, Next.js, React Flow, Gemini, LangChain, MongoDB

- Built VS Code extension with **TypeScript** for inline cost annotations, heatmaps, and one-click optimizations.
- Indexed **10,000+**-line codebases in under **3 seconds** using **AST**-based inefficiency detection.
- Cut developer API spend by up to **40%** by providing real-time API cost insights via regex and usage caching.
- Created a React Flow sandbox for budget-aware system planning with **RAG Gemini** chatbot.

### Nebula AI | React Native, FastAPI, PostgreSQL, Docker, GCP

- Shipped a mobile app that turns handwritten notes into a searchable, **LaTeX**-formatted AI knowledge base.
- Enabled semantic search over **1,536-dim embeddings**, cutting retrieval time by **40%** with IVFFlat indexing.
- Implemented an **OCR+AI pipeline** with Gemini/OpenAI, enabling text extraction and AI-driven summaries.
- Containerized backend with **Docker**, deployed to **Google Cloud Run** with automated CI/CD pipelines.

## COMMUNITY

### Hack the North

*Logistics Organizer*

Feb 2026 – Present

Waterloo, Ontario

- Led end-to-end operations for **300+ volunteers** supporting Hack the North, a **1,000+**-participant hackathon.
- Redesigned the volunteer shift-selection system to prevent assignment race conditions and streamline sign-up.
- Oversaw **300+** 3D-printed keychains from design to production given time and material constraints.

## TECHNICAL SKILLS

**Hardware:** Verilog, RISC-V, FPGA, Altera Quartus Prime, Fusion 360, Arduino, Teensy

**Software:** C/C++, Python, Bash, Numpy, TypeScript, JavaScript, HTML/CSS, SQL, Java, R, React, FastAPI, Next.js

**Tools:** Linux/Unix, Git(hub), Docker, GCP, AWS, Azure, PostgreSQL, MongoDB, Figma, Jira/Confluence